

Does a small VLA listen, and where does it break?

Graded robustness of SmolVLA on LIBERO and a residual RL correction

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Abstract

Vision-language-action (VLA) models are evaluated almost exclusively in the conditions they were trained on. We probe SmolVLA [11], a 450M-parameter open VLA, on the LIBERO benchmark [7] with three questions. *Does it listen?* In LIBERO-Goal, where ten tasks share one scene, we swap instructions and check all ten goal predicates at every step: the robot reaches the goal of the instruction it is given in 73% of swapped episodes and the environment’s own goal in 0%, but hand-written paraphrases of the same instruction succeed far less often (23% against 80%), a gap that retrieving the closest training instruction closes. *How does it break?* Under five graded perturbations, success declines gradually with camera viewpoint, collapses past sharp thresholds for lighting and pixel noise, and is halved by an offset of about 3° per arm joint at the start of the episode; combinations of perturbations that are each harmless make 71% of held-out episodes fail. *Can a small corrector help?* A residual policy trained with PPO on top of the frozen VLA improves success on the initial states it was trained on but not on held-out ones, and mostly learns to finish successful episodes faster. All results come from 3 435 simulated episodes run on a laptop, with per-episode logs and code released.

1 Introduction

VLA models map a camera image and a sentence to robot actions with a single network. Benchmark scores on LIBERO are now close to saturation, which says little about what happens when the scene moves slightly away from the demonstrations. Recent robustness suites [3, 15] answer this with many perturbation types at a few fixed levels and report that models are fragile to viewpoint and initial state, and largely ignore language.

This report looks at a narrower set of questions in more depth, on one small model:

- **Listening.** If the scene is held fixed and only the instruction changes, does the robot change what it does? Checking every task’s goal, not only the one the environment scores, separates “ignores the instruction” from “follows it but fails”.
- **Shape of the degradation.** Each perturbation has a continuous intensity; we fit a dose-response curve and estimate the intensity x_{50} at which success falls to half its unperturbed value.
- **Repair without retraining.** A small policy trained with PPO adds a bounded correction to the frozen VLA’s actions (residual RL [5, 12]).

2 Related work

Robustness of VLAs. LIBERO-Plus [3] perturbs LIBERO along seven axes (camera, lighting, language, sensor noise, robot initial state, layout, textures) and finds camera and initial-state changes the most harmful, and language variations almost harmless, which the authors read as models ignoring language. LIBERO-PRO [15] reports that success can collapse under object and position changes and argues that high scores reflect memorisation. We reuse their perturbation axes but sweep intensity continuously, and our listening test scores every goal of the shared scene rather than only the environment’s own goal.

Residual policies. Residual RL learns a correction on top of a fixed controller or policy [5, 12]; ResiP [1] and Policy Decorator [14] apply it to imitation-learned diffusion or transformer policies. We apply the same idea to a VLA, targeting one specific robustness failure.

3 Setup

Model and simulator. We use the Hub checkpoint `lerobot/smolvla_libero` (revision 31d453f), evaluated through LeRobot [2] 0.6.1 processors so that normalisation, tokenisation and camera naming match `lerobot-eval`. The policy predicts chunks of 50 relative end-effector actions with flow matching [6] and executes the whole chunk before re-planning. Images are 360×360 from the third-person (*agentview*) and wrist cameras. Everything ran on an Apple M1 laptop (8 GB, PyTorch MPS, bf16 weights); LIBERO runs natively on macOS once an unused Linux-only dependency is skipped.

Protocol. Episode e of a task always starts from LIBERO initial state e with seed e (which also fixes the flow-matching noise), whatever the condition, so conditions are compared on identical episodes. Hard resets are used throughout. Success rates are reported with Wilson 95% intervals [13]; paired comparisons use an exact McNemar test.

Perturbations (Fig. 1). *Camera orbit:* the agentview camera rotates about the vertical axis through the point it looks at on the table, $0\text{--}60^\circ$, sign alternating with the episode; the wrist camera is untouched. *Arm initial pose:* after reset, each of the seven joints is shifted by $\pm\delta$, $\delta \in [0, 0.2]$ rad, with signs drawn per episode; objects are not moved. *Lighting:* every light in the scene is scaled down by up to 90%. *Pixel noise:* Gaussian noise of standard deviation up to 60 grey levels on both cameras. At the strongest level all task-relevant objects remain visible (Fig. 1), except at 90% light removal where the scene is very dark but still readable.

Dose-response fit. For each perturbation we fit $p(x) = s_0 / (1 + e^{(x-x_{50})/w})$ by binomial maximum likelihood, with s_0

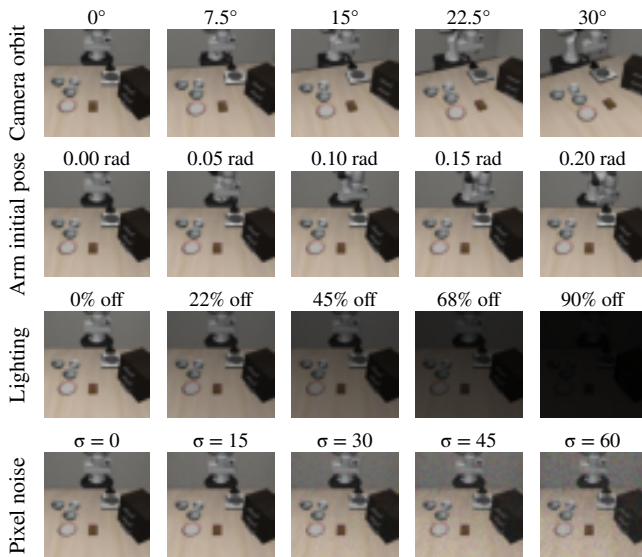


Figure 1: What the policy sees from the main camera at intensities 0, 0.25, 0.5, 0.75 and 1 (LIBERO-Spatial task 0, initial state 0). The arm offset also moves the wrist view (not shown).

fixed to the unperturbed success rate on the same episodes, so that x_{50} is exactly the intensity at which success is halved. Its 95% interval comes from a bootstrap over tasks (whole tasks resampled), since task difficulty varies far more than episode outcomes within a task.

4 Results

Baseline. Without perturbation SmolVLA succeeds in 59/100 (59%, CI 49–68) LIBERO-Spatial episodes (10 tasks $\times$ 10) and in 84/100 (84%, CI 76–90) LIBERO-Goal episodes. The Spatial rate is well below the 90% reported in [11]. To rule out our own evaluation loop, we re-ran two tasks with the official lerobot-eval: 14/20 (70%), against 12/20 (60%) with our loop on the same tasks, a difference well within sampling noise. Two tasks (5 and 8) are solved in at most 2 of 10 episodes. All conclusions below are relative to this measured baseline.

4.1 SmolVLA follows the instruction

Fig. 2a is the matrix one would get from the environment alone: success only on the diagonal. Read in isolation it is ambiguous, since a robot that ignores language would also fail off-diagonal if it performed some fixed behaviour unrelated to goal j . Panel (b) removes the ambiguity. When told to do task i inside the environment of task j , the robot reaches goal i in 73% of the 270 swapped episodes, close to the 25/30 (83%, CI 66–93) obtained with matched instructions, and reaches the environment’s goal in 0/270 of them; only 2/270 episodes complete a third, unrelated goal. The initial states of the ten tasks differ slightly (each task has its own set), so the robot is not keyed on the start configuration either. The weakest instructions are those whose objects are close to other objects (task 3, bowl into the top drawer; task 6, cream cheese into the bowl).

This contrasts with the conclusion of LIBERO-Plus that VLAs largely ignore language. A swap test is a direct check:

in a fixed scene, only the instruction can tell the robot which of ten goals to pursue.

4.2 ... but to the exact words

The same model is brittle to rewording (Fig. 3). Hand-written paraphrases that keep the goal unchanged succeed in only 35/150 (23%, CI 17–31) of episodes, against 40/50 (80%, CI 67–89) with the original wording on the same episodes, and the rate drops with the distance from the training wording: 20/50 for a close paraphrase (“*place the bowl on the stove*”), 11/50 for a reworded one, 4/50 for the most distant. Small changes matter: “*turn the stove on*” works, “*switch on the stove*” does not. An empty instruction and an unrelated one (“*sing a song*”) both give 0/50. In none of the 215 failed episodes does the robot complete any of the ten goals: without an instruction it recognises, it does not fall back on a default task, it does nothing useful.

Taken together, the instruction is what selects the behaviour, but it acts more like a key into a small set of learned behaviours than like language that is understood. For this model and suite we therefore find sensitivity to language on both counts: naming another goal changes the behaviour, and rewording the same goal breaks it. Model, suite and perturbation types all differ from LIBERO-Plus, and we did not reproduce their setting, so this is a statement about SmolVLA on LIBERO-Goal only.

A fix at the input. Mapping each instruction to the closest of the 40 LIBERO training instructions with a sentence encoder (all-MiniLM-L6-v2 [9]; below a cosine similarity of 0.5, fixed beforehand, the instruction is left unchanged) brings paraphrases from 35/150 (23%, CI 17–31) to 120/150 (80%, CI 73–86) on LIBERO-Goal, the level of the original wording, and rejects “*sing a song*”. On LIBERO-Spatial, where instructions differ only by a spatial relation, the encoder confuses *next to* and *on* for 5 of 20 paraphrases and success only rises from 26/100 (26%, CI 18–35) to 54/100 (54%, CI 44–63).

4.3 Different ways of breaking

The perturbations do not just differ in strength; the curves have different shapes (Fig. 4). We report two estimates of the breaking point: the logistic x_{50} , and a model-free one (pooled success made non-increasing by isotonic regression, then linearly interpolated at half the baseline), both with task-bootstrap intervals.

- **Camera orbit: gradual.** Success declines steadily with the angle, with no plateau and no cliff: it is halved around $x_{50} = 40^\circ$ [25, 61] and is still above zero at 60° , a viewpoint far from anything in the demonstrations. The decline is strongly task dependent: some tasks are unaffected at the largest angle, others fail from the first level. The two orbit directions are not equally harmful at intermediate angles, but direction is confounded with the initial state here (it alternates with the episode index), so we do not interpret it.
- **Arm initial pose: narrow tolerance, then a cliff.** Offsets up to 0.02 rad per joint ($\approx 1^\circ$) leave success unchanged; between 0.03 and 0.05 rad it is halved (model-free $x_{50} = 0.048$ rad [0.031, 0.082]; logistic 0.078 rad), and a long tail follows. The objects have not moved and the wrist camera still sees

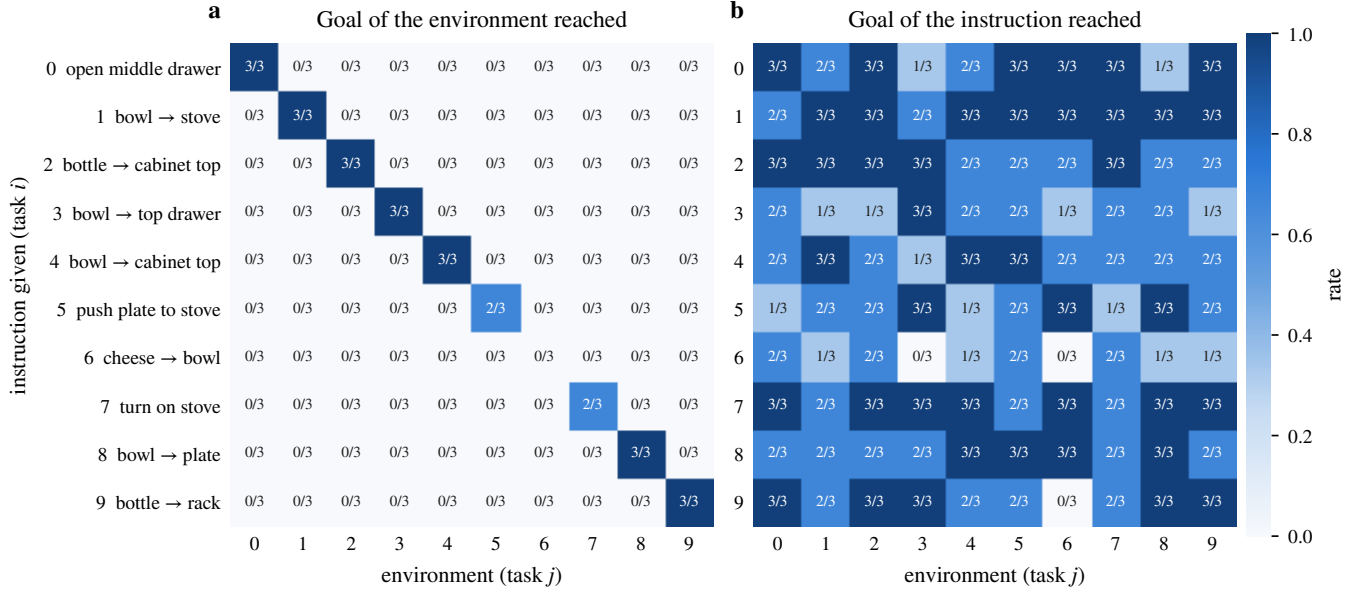


Figure 2: Listening test on LIBERO-Goal (3 episodes per cell). The environment of task j is built and the instruction of task i is given. (a) Success as scored by the environment (goal j). (b) Whether the goal of the *given* instruction i was reached at any point of the episode, obtained by evaluating all ten goal predicates of the shared scene at every step. Off the diagonal the robot never completes the environment’s own task, but reaches the instructed goal in 197/270 (73%, CI 67–78) of episodes.

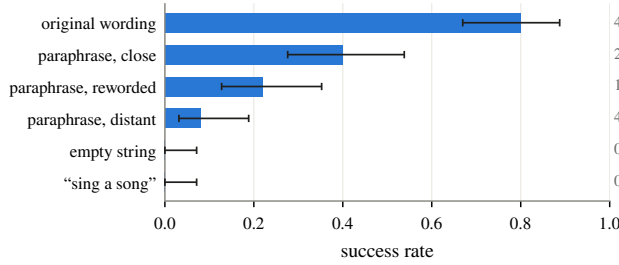


Figure 3: Instruction variants on LIBERO-Goal (environment and instruction of the same task, 5 episodes per task). Paraphrases were written by hand, from close to the original wording to fully reworded. Wilson 95% intervals.

them; the policy only starts from an arm configuration a few degrees away from the demonstrations.

- **Lighting: threshold.** Removing up to 67.5% of the light has no measurable effect; success is then halved at $x_{50} = 0.74$ [0.72, 0.78] of light removed and falls below a quarter of the baseline at 81%. A single fixed-level test would either miss the effect entirely or report a collapse.
- **Pixel noise: threshold.** No effect at a standard deviation of 15 grey levels, a small one at 30, then a collapse: success is halved at $x_{50} = 42$ [35, 47] and is close to zero at 60, although the objects remain easy to see for a human at that level (Fig. 1).

Gradual decline for viewpoint, sharp thresholds for photometric changes (lighting, noise) and a narrow tolerance for the arm configuration: one fixed perturbation level per axis would have placed these on a single “robust/fragile” scale and missed the differences. **Both cameras are needed.** Blacking out one stream entirely drops success from 33/50 to 5/50 without

the main camera and 6/50 without the wrist camera (same episodes). The policy cannot do without the third-person view, yet tolerates rotating it by 20–30° with a moderate loss, which suggests it relies on coarse scene layout from that view rather than on precise image coordinates.

Harmless alone, harmful together. We restrict each perturbation to a range where it alone costs at most about 15% of the baseline (camera $\leq 7.5^\circ$, arm ≤ 0.03 rad, light removed $\leq 67.5\%$, noise $\sigma \leq 30$) and keep the 15 episodes that succeed under each one alone at the top of its range. A MAP-Elites search [8] over their combinations (120 combinations, four search episodes each) finds failures almost everywhere: 98/120 combinations fail at least one episode, the full combination fails 4/4, and the six most harmful ones make 47/66 (71%, CI 59–81) of the 11 held-out episodes fail. One-factor-at-a-time curves overestimate the margin when several conditions drift together.

4.4 A residual corrector learns, but does not generalise

Target and design. The arm-offset weakness is a natural target: the perturbation is visible in proprioception, so a corrector without images can in principle detect it. We use LIBERO-Spatial task 2 at 0.05 rad per joint, where the VLA alone succeeds in about half of the episodes. The corrector is an MLP (two hidden layers of 128) that sees end-effector position and orientation, gripper opening, the VLA’s proposed action and the episode progress, and outputs $\Delta a \in [-1, 1]^7$; the executed action is $\text{clip}(a_{\text{VLA}} + 0.2 \Delta a)$. Its last layer is initialised to zero, so before training it reproduces the VLA exactly; we checked this on full episodes (identical outcomes and episode lengths). It is trained with PPO [10] in a single-file implementation in the style of CleanRL [4], with a sparse reward (1 on success), $\gamma = 0.995$, four simulators in one

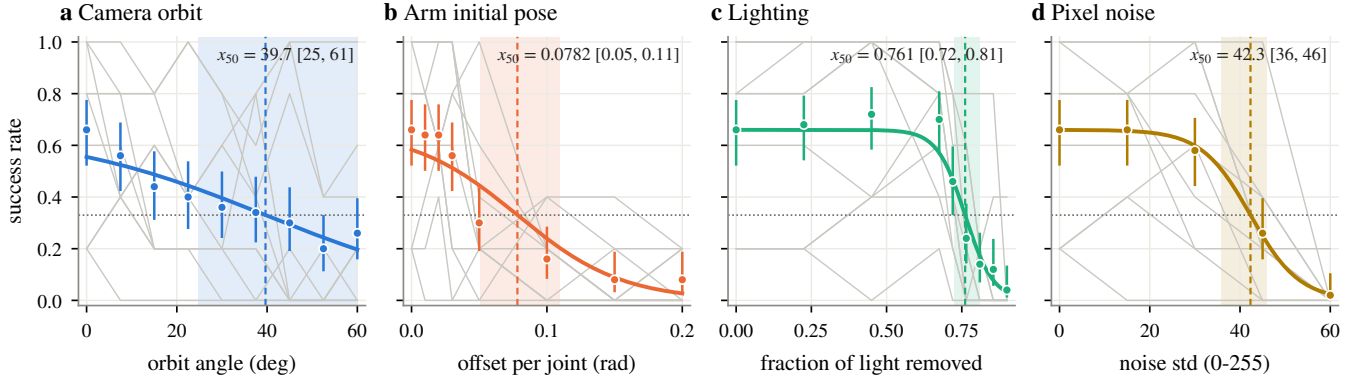


Figure 4: Success versus perturbation intensity on LIBERO-Spatial (10 tasks, 5 episodes per task and level; the unperturbed point uses the same episodes). Dots: pooled success with Wilson 95% intervals. Grey: individual tasks. Line: logistic fit with s_0 fixed to the unperturbed rate; dotted: half of it. Dashed line and band: breaking point x_{50} and its 95% task-bootstrap interval.

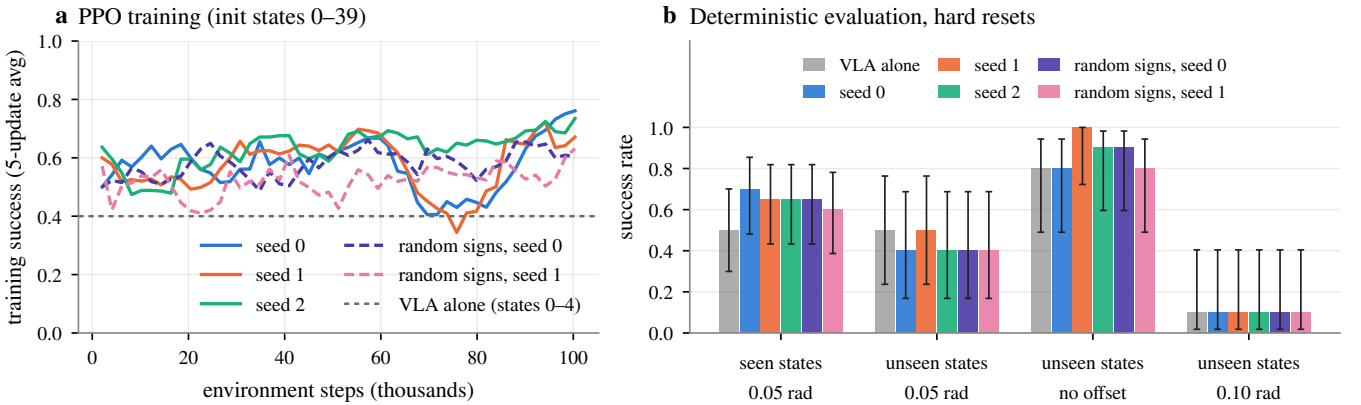


Figure 5: Residual RL on the arm-offset weakness (LIBERO-Spatial task 2, offset 0.05 rad per joint). (a) Success of the stochastic policy during PPO, moving average over 5 updates (about 60 episodes). (b) Deterministic evaluation with hard resets, on initial states seen during training (0–19, 20 episodes) and on held-out ones (40–49, 10 episodes per bar). Wilson 95% intervals.

process sharing batched VLA calls, and 100k environment steps per run (about 400 episodes, 3 h on the laptop at 10 steps/s). Training uses initial states 0–39; states 40–49 are kept for evaluation.

Results (Fig. 5). Training success rises modestly (from 0.60, 0.51, 0.54 to 0.70, 0.66, 0.71 over the first and last ten updates of each of the 3 seeds); seeds 0 and 1 dip around 70k steps before recovering, seed 2 does not. On initial states seen in training, the deterministic corrector succeeds in 14/20, 13/20, 13/20 episodes against 10/20 for the VLA alone, and every discordant episode favours the corrector (McNemar $p = 0.12$, 0.38, 0.38, not significant at this sample size). On held-out initial states the gain disappears: 4/10, 5/10, 4/10 against 5/10 at the training level, 1/10, 1/10, 1/10 against 1/10 at 0.10 rad, and 8/10, 10/10, 9/10 against 8/10 without offset (so the corrector does not hurt the unperturbed task).

What does transfer is speed: successful episodes are shorter with the corrector, from 110 to 88–91 steps on seen states and from 95 to 88–92 on held-out ones. This is what the objective rewards most directly: with discounting a faster success is worth more, and speeding up an episode that already works is observed in every rollout, whereas recovering a failure is a rare event in 400 episodes. The recoveries that do happen are

tied to the configurations seen in training; with 40 of them, the corrector has little to interpolate from.

Drawing new offset signs at every training reset, to expose the corrector to more varied starting poses, gives the same pattern (13/20, 12/20 on seen and 4/10, 4/10 on held-out initial states), so the variety of arm offsets is not what limits generalisation.

A training-free alternative, executing 10 instead of 50 actions per chunk so that the VLA re-plans five times more often, gives 10/10, 4/10 and 3/10 on the same held-out episodes (no offset, 0.05 and 0.10 rad), not distinguishable from the VLA alone with 10 episodes, at 2.4× the cost.

5 Limitations

Everything is in simulation, on one model and one checkpoint, with 3–10 episodes per cell; intervals are correspondingly wide and per-task curves noisy. Our baseline is well below the numbers reported in the SmolVLA paper (which may not use this Hub checkpoint), so all conclusions are relative to the measured baseline. Inference ran on Apple MPS in bf16; a GPU run with the same configs is provided but was not used for the reported numbers. The residual corrector sees proprioception only, which is why we target a perturbation that

is visible in proprioception.

6 Conclusion

On LIBERO, SmolVLA is neither deaf to language nor robust to it: the instruction decides which task is attempted, but only in wordings close to the training ones. Its failures under distribution shift have distinct shapes, gradual for viewpoint and abrupt for photometric changes and arm configuration, which a single perturbation level per axis would hide, and perturbations harmless one at a time fail together. Looking up known wordings repairs most paraphrases. A residual corrector trained with a few hundred episodes does not repair the arm-pose weakness on unseen configurations, even with more diverse arm offsets during training. A natural next step is the same protocol on larger VLAs.

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